Introduction

System: A system is an organized set of components that are connected and work together to achieve a specific purpose.

Examples

- A car – engine, wheels, brakes, and steering all work together to move safely.
- A human body – heart, lungs, and brain function together to keep a person alive.

System Objectives

Every system is created to fulfill a specific purpose. These purposes help us understand why a system exists and how it functions.

1. Information Processing

Purpose: To collect, store, process, and share information in a useful form.

Examples:

- Computer system – processes user data (like calculations, files, images) into meaningful outputs such as reports, documents, or results.
- Human brain – processes sensory data from eyes, ears, and other senses to understand the surroundings.

2. Supporting Other Systems

Purpose: To act as a platform, resource, or foundation for other systems to work effectively.

Examples:

- Mobile phone – provides a platform to run applications like WhatsApp, games, or banking apps.
- Sun – provides energy that supports ecosystems, plants (photosynthesis), and all forms of life on Earth.

3. Achieving Specific Goals

Purpose: To perform a particular task or complete a defined process.

Examples:

- Thermostat system – maintains room temperature at a desired level.
- Car engine system – converts fuel into mechanical energy to move the vehicle.

System Components

Components are the building blocks of a system. Each has a specific role and contributes to the overall working of the system.

Examples:

- In a computer system: CPU, memory, input/output devices.
- In the human body: heart, lungs, blood vessels, and brain.

System Environment

The environment of a system means everything outside the system that affects it. The environment gives inputs to the system, and the system gives outputs back to the environment.

Examples:

- A computer takes input from a keyboard (environment) and gives output to a screen.
- A plant takes sunlight and water (environment) and gives oxygen back.

1. Static Environment

A static environment does not change on its own; it only changes when the system acts upon it.

- A library system – books stay in place unless borrowed or returned.
- A traffic light system – changes follow fixed programmed rules.
- A school timetable – stays the same unless updated by the administration.
- A parking lot system – space status changes only when cars enter or leave.

2. Dynamic Environment

A dynamic environment changes independently of the system.

- A stock market system – share prices fluctuate constantly.
- Weather conditions – temperature, rainfall, and wind change unpredictably.
- A social media platform – new posts, comments, and trends appear anytime.
- A hospital emergency system – number and type of patients change unexpectedly.

3. Deterministic Environment

In a deterministic environment, the effect of a system's output is certain and predictable.

- A calculator – solving 2+2 will always give 4.
- A vending machine – inserting money always delivers the selected product.
- A train schedule – trains run at fixed times (under normal conditions).
- A digital clock – always moves forward in predictable intervals.

4. Non-deterministic Environment

In a non-deterministic environment, the system's outputs may be uncertain, random, or based on probability.

- A lottery system – the winner is chosen randomly.
- A weather forecasting system – predictions are probability-based.
- A traffic jam situation – exact waiting time cannot be predicted.
- A job recruitment process – many unpredictable factors affect selection.

communication

Communication of a system means the exchange of information between different components of a system. These components interact with each other to achieve the system's objective.

Example:

The CPU's objective is to process data. To do this, it communicates with RAM by fetching data and instructions. After processing, it may send results back.

System Interaction with Environment

Systems constantly interact with their environment through inputs (what they receive) and outputs (what they produce).

Examples:

- A weather monitoring system collects sensor data (input) and produces forecasts (output).
- A computer system interacts with printers and scanners (input/output devices).
- A biological system: plants absorb sunlight (input) and release oxygen (output).